1. Sum of Even and Odd Numbers

```
n = int(input("Enter a number: "))
even_sum = 0
odd_sum = 0

for i in range(1, n + 1):
    if i % 2 == 0:
        even_sum = even_sum + i
    else:
        odd_sum = odd_sum + i

print("Even sum:", even_sum)
print("Odd sum:", odd_sum)
```

2. Count Digits in a Number

```
n = abs(int(input("Enter a number: ")))
count = 0

if n == 0:
    count = 1
else:
    for i in range(n):
        count = count + 1
        n = n // 10
        if n == 0:
            break

print("Digits:", count)
```

3. Find the Second-Largest Number

```
numbers = [10, 25, 8, 40, 15]

largest = numbers[0]
second = numbers[1]

if second > largest:
    temp = largest
    largest = second
    second = temp

for i in range(2, len(numbers)):
    if numbers[i] > largest:
        second = largest
        largest = numbers[i]
    elif numbers[i] > second:
        second = numbers[i]

print("Second largest:", second)
```

4. Count Vowels and Consonants

```
s = input("Enter a string: ")
vowels = 0
consonants = 0

for ch in s.lower():
    if ch.isalpha():
        if ch in "aeiou":
            vowels = vowels + 1
        else:
            consonants = consonants + 1

print("Vowels:", vowels)
print("Consonants:", consonants)
```

5. Calculate Factorial

```
n = int(input("Enter a positive number: "))
fact = 1

for i in range(1, n + 1):
    fact = fact * i

print("Factorial:", fact)
```

6. Numbers Divisible by 7 but Not by 5

```
count = 0

for i in range(1, 201):
    if i % 7 == 0 and i % 5 != 0:
        print(i, end=" ")
        count = count + 1

print()
print("Total:", count)
```

7. Calculate Electricity Bill

```
units = []

for i in range(5):
    u = int(input("Enter units: "))
    units.append(u)

for u in units:
    if u <= 100:
        bill = u * 2
    elif u <= 200:
        bill = 100 * 2 + (u - 100) * 3
    else:
        bill = 100 * 2 + 100 * 3 + (u - 200) * 5

    print("Units:", u, "Bill: Rs.", bill)
```

8. Reverse a Number

```
n = abs(int(input("Enter a number: ")))
reverse = 0

for i in range(n):
    digit = n % 10
    reverse = reverse * 10 + digit
    n = n // 10
    if n == 0:
        break

print("Reversed number:", reverse)
```

9. Analyze Student Marks

```
marks = [45, 80, 35, 90, 75, 60, 25]
passed = 0
failed = 0
distinction = 0

for m in marks:
    if m >= 50:
        passed = passed + 1
    else:
        failed = failed + 1

    if m >= 75:
        distinction = distinction + 1

print("Passed:", passed)
print("Failed:", failed)
print("Distinction:", distinction)
```

10. Perfect Numbers from 1 to 1000

```
for n in range(1, 1001):
    total = 0
    for i in range(1, n):
        if n % i == 0:
            total = total + i

    if total == n:
        print(n)
```